

Logarithms and Their Properties



Evaluating logarithms

1 Evaluate each logarithm exactly:

a $\log_2 32$

b $\log_3 \frac{1}{9}$

c $\log 1000$

d $\ln e^7$

2 Use the change-of-base formula to evaluate $\log_5 18$, to three decimal places.

3 State the domain of $f(x) = \log(x - 4)$ and the inverse function of $g(x) = 2^x$.

Properties of logarithms

4 Expand fully using properties of logarithms: $\ln\left(\frac{x^3\sqrt{y}}{z^2}\right)$.

5 Write as a single logarithm: $2\log x + 3\log y - \log z$.

6 Given $\log_b 2 = 0.431$ and $\log_b 3 = 0.683$, evaluate:

a $\log_b 6$

b $\log_b 8$

c $\log_b 1.5$

Graphs of logarithmic functions

7 For $y = \log_2(x + 4) - 1$, state:

a the domain

b the vertical asymptote

c the x-intercept

8 Describe how the graph of $g(x) = -\ln(x - 2) + 3$ is obtained from $y = \ln x$.

Logarithmic scales and applications

9 The Richter scale measures earthquake magnitude by $M = \log\left(\frac{I}{I_0}\right)$, where I is intensity and I_0 is a reference intensity.

a An earthquake has $M = 6$. Express I in terms of I_0 .

b How many times more intense is a magnitude-7 earthquake than a magnitude-5 earthquake?

10 Sound intensity level in decibels is $\beta = 10\log\left(\frac{I}{I_0}\right)$. If a sound's intensity doubles, by how much does β increase, to two decimal places?

More practice with logarithm properties

11 Expand fully: $\log_2 \left(\frac{8x^2}{\sqrt[3]{y}} \right)$.

12 Write as a single logarithm and simplify: $\frac{1}{2} \ln 16 + \ln 3 - \ln 2$.